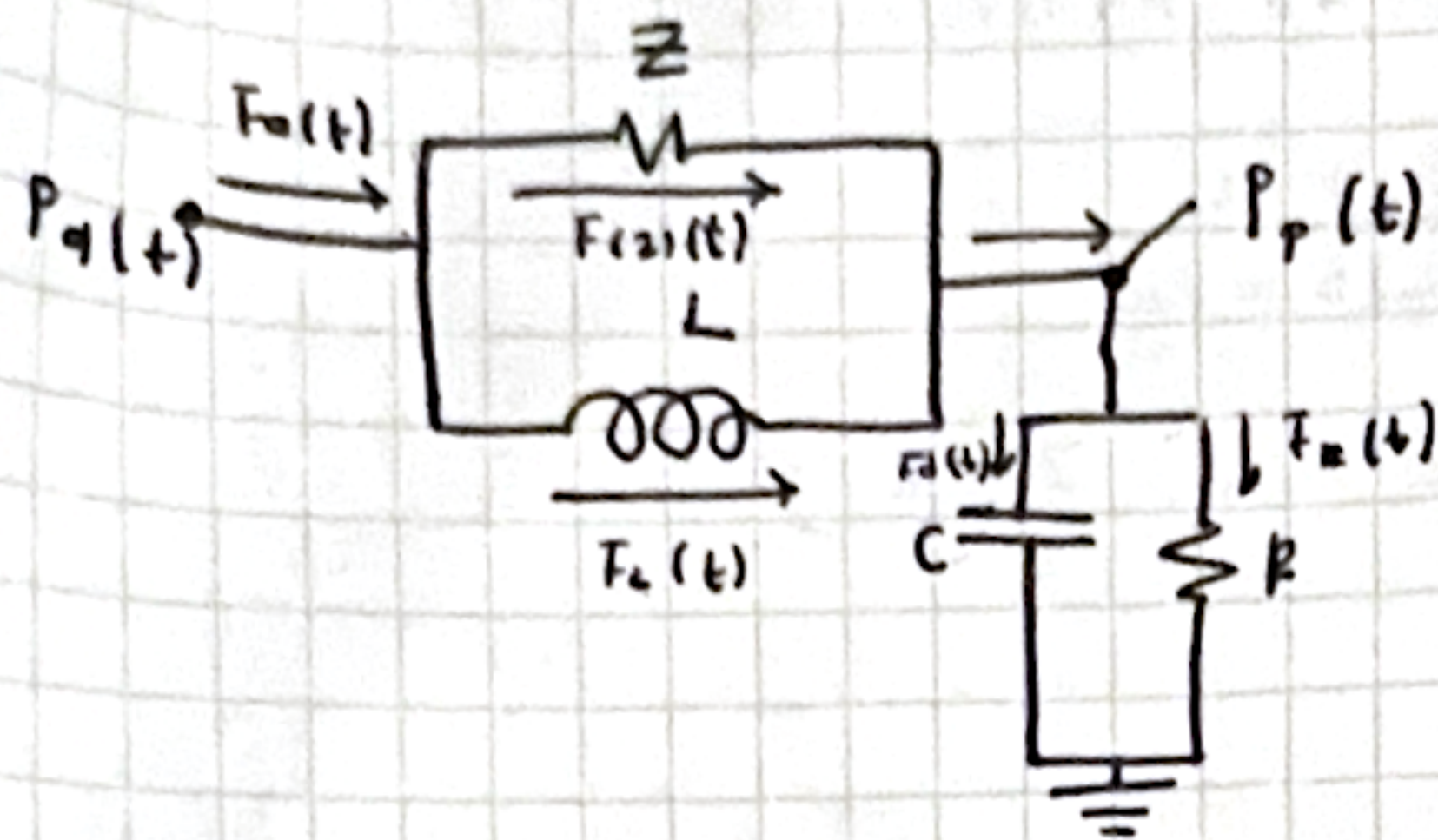


PRACTICA 5.4: SISTEMA CARDIOVASCULAR



ecuación principal:

$$F_a(t) = F_a(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_a(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_C(t) = \frac{dP_p(t)}{dt}$$

$$F_L(t) = \frac{1}{L} \int [P_a(t) - P_p(t)] dt \quad F_R(t) = \frac{P_p(t)}{R}$$

procedimiento algebraico

$$\frac{P_a(t)}{Z} - \frac{P_p(t)}{Z} + \frac{1}{L} \int [P_a(t) - P_p(t)] dt = C \frac{dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{LS} = CS P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{LS} \right) P_a(s) = \left(CS + \frac{1}{R} + \frac{1}{Z} + \frac{1}{LS} \right) P_p(s)$$

$$\frac{LS + Z}{ZLS} P_a(s) = \left(\frac{CLS^2Z + LZS + RLS + RZ}{RZLS} \right) P_p(s)$$

$$\frac{P_p(s)}{P_a(s)} = \frac{LS + Z}{[CLRZ]s^2 + [LZ + LR]s + RZ}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRS + RZ}{[CLRZ]s^2 + [LZ + LR]s + RZ} \rightarrow \text{Función de transferencia}$$

Estabilidad en lazo abierto

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda_{1,2} = \frac{-(Lz + LR) \pm \sqrt{(Lz + LR)^2 - 4(LR^2z)}}{2CLRz}$$

$$a = (LRz)$$

$$b = (Lz + LR)$$

$$c = R^2z$$

$$\lambda_1 = p_c < 0$$

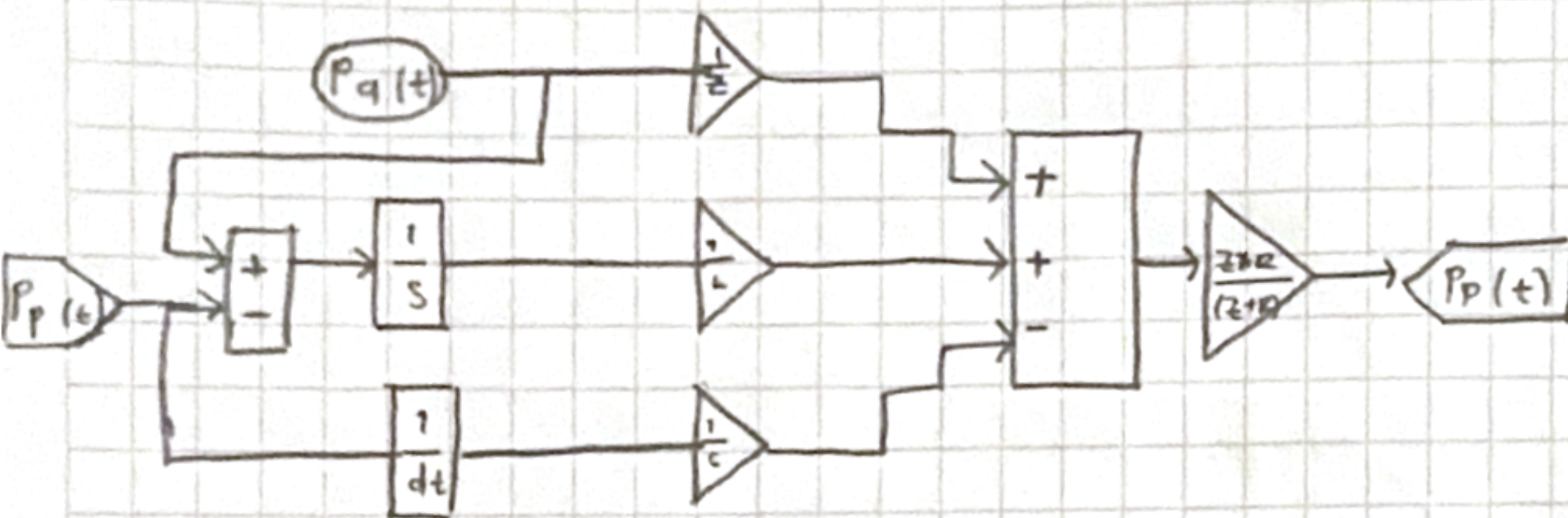
$$\lambda_2 = p_c < 0$$

La respuesta del sistema es estable

Modelo de ecuaciones integro diferenciales

$$P_p(t) \left(\frac{1}{R} + \frac{1}{Z} \right) = \frac{P_a(t)}{Z} + \frac{1}{L} \int [P_a(t) - P_p(t)] dt - \frac{dP_p(t)}{dt}$$

$$P_p(t) = \left(\frac{P_a(t)}{Z} + \frac{1}{L} \int [P_a(t) - P_p(t)] dt - \frac{dP_p(t)}{dt} \right) \frac{ZR}{Z+R}$$



Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s P_a(s) \left[1 - \frac{P_p(s)}{P_a(s)} \right]$$

$$= \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \left[1 - \frac{RLs + RZ}{CLRZs^2 + [LZ + LR]s + RZ} \right]$$

$$= 0V$$